

Network Case Study Summary (Study Case 1 & 2)

Study Case 1: Multi-LAN Network with Two Switches

Topology Overview: Two switches are connected to form a larger LAN. Multiple PCs are grouped into segments (A, B, C), all connected through Switch 1 and Switch 2.

Step-by-Step Configuration:

1. Place devices: Add PCs and two switches in Cisco Packet Tracer.
2. Connect cables: Use straight-through cables to connect PCs to switches and connect both switches together.
3. Assign IP addresses: Each PC should have a unique IP within the same subnet (e.g., 192.168.1.x).
4. Configure subnet mask: Use 255.255.255.0 for all devices.
5. Verify connectivity: Use ping between PCs across both switches.
6. Troubleshooting: Check cable connections, IP conflicts, and switch ports if ping fails.

Expected Result:

All PCs across both switches should successfully communicate with each other.

Study Case 2: Network with Server, Web Server, and Wireless Access

Topology Overview: A central switch connects PCs, a main server, a web server, and an access point. A laptop connects wirelessly through the access point.

Step-by-Step Configuration:

1. Place devices: Add PCs, servers, one switch, and an access point.
2. Connect wired devices: Use straight-through cables to connect PCs and servers to the switch.
3. Configure servers: Assign IP addresses and enable services (e.g., HTTP for web server).
4. Configure access point: Set SSID and security settings.

5. Connect laptop: Use wireless settings to connect to the access point.
6. Assign IP to laptop: Ensure it is in the same network (via DHCP or static IP).
7. Test connectivity: Ping servers and access web server from laptop.
8. Troubleshooting: Check wireless connection, IP settings, and server services.

Expected Result:

All wired and wireless devices should communicate successfully, and the laptop should access the web server.